==

INTRUSION DETECTION SYSTEM

Network Malicious Detection (Benign vs Malicious)

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All BOT-IOT Dataset shape: (972905, 22)

Dataset after drop duplicates: (972905, 22)

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Label (Benign and Malicious) counts:

attack

1   971149

0   1756

Name: count, dtype: int64

Features after cleaning: 21

Data split: Train=681033 (70%), Test=291872 (30%)

Training model...

Training time: 11.61 seconds

Cross-Validation (on training set)...

10-Fold CV Accuracy on Train: 0.9999 ± 0.0000

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PERFORMANCE SUMMARY

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Accuracy | Train: 100.00% | Test: 99.99%

Balanced Acc | Train: 100.00% | Test: 99.62%

Precision | Train: 100.00% | Test: 99.99%

Recall | Train: 100.00% | Test: 99.99%

F1 | Train: 100.00% | Test: 99.99%

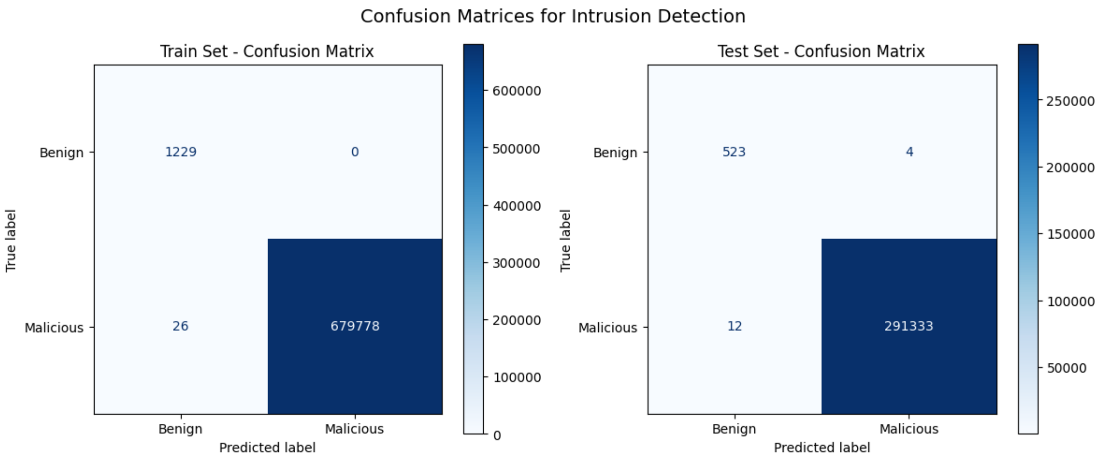
MCC | Train: 98.96% | Test: 98.49%

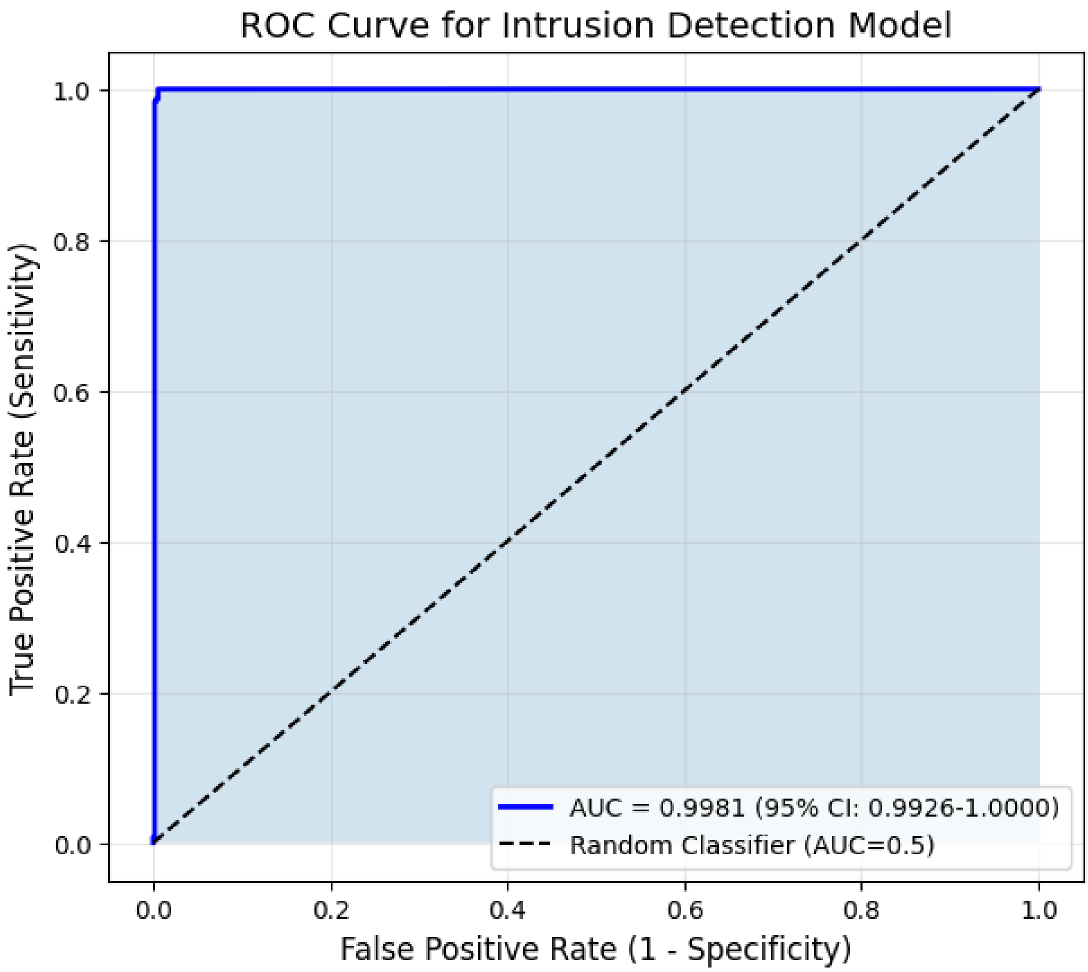
Kappa | Train: 98.95% | Test: 98.49%

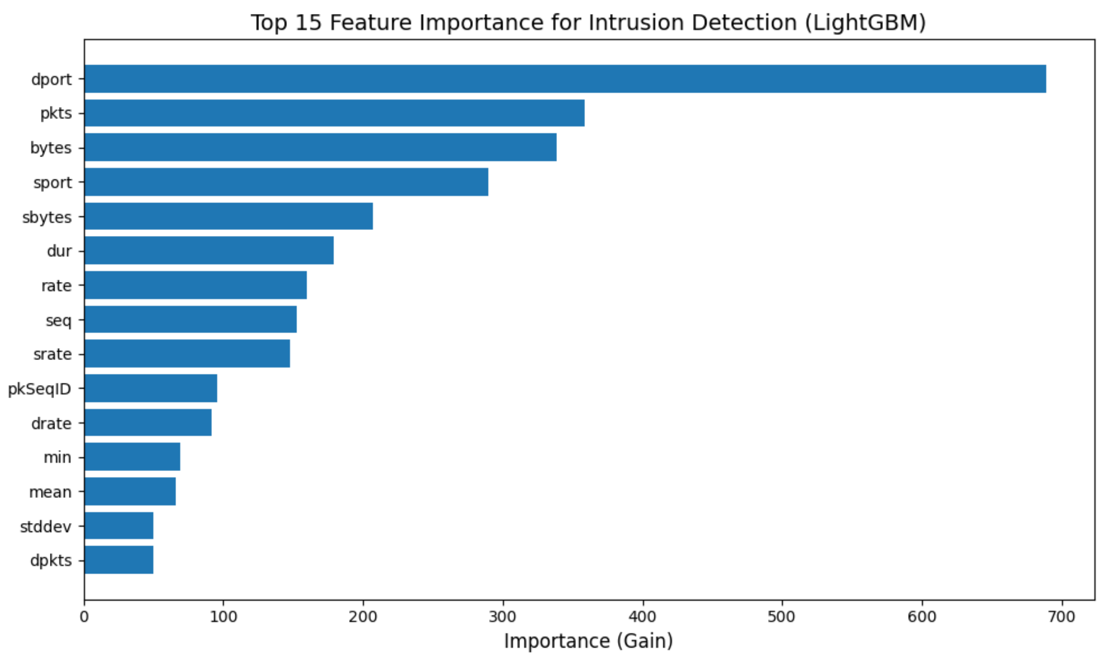
Specificity | Train: 100.00% | Test: 99.24%

Sensitivity | Train: 100.00% | Test: 100.00%

AUC | Train: 0.00% | Test: 99.81%







Top 10 features:

Feature Importance

3	dport	689
4	pkts	359
5	bytes	339
2	sport	290
16	sbytes	207
8	dur	179
18	rate	160
7	seq	153
19	srate	148
0	pkSeqID	96

SHAP analysis (sampling 100 test samples)...

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FEATURE IMPORTANCE BASED ON SHAP

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Feature	Strength	Impact %
sbytes	0.1600	33.56
rate	0.0791	16.59
pkts	0.0718	15.05
dport	0.0529	11.10
srate	0.0332	6.97
dur	0.0157	3.30
drate	0.0141	2.96
bytes	0.0120	2.52
dpkts	0.0076	1.60

seq	0.0071	1.50
stddev	0.0052	1.09
spkts	0.0047	0.98
mean	0.0042	0.89
sport	0.0041	0.86
min	0.0024	0.50

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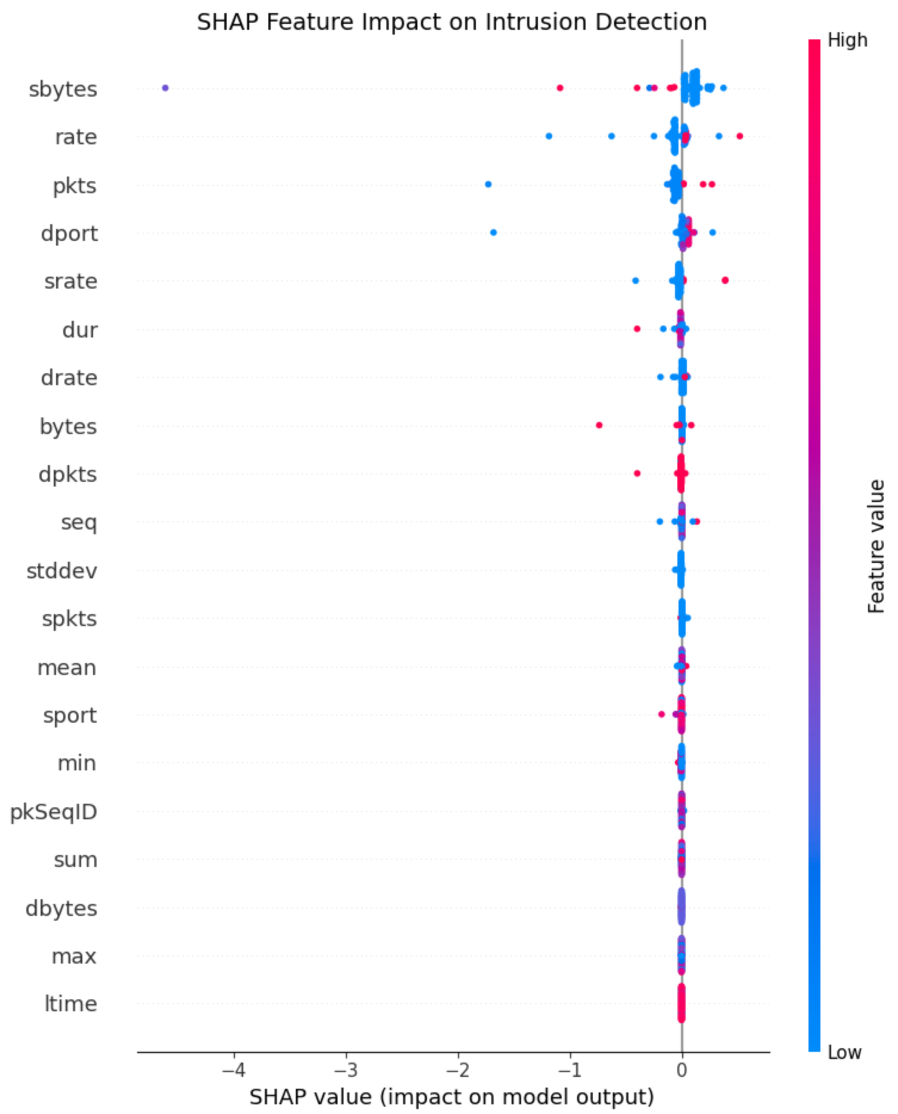
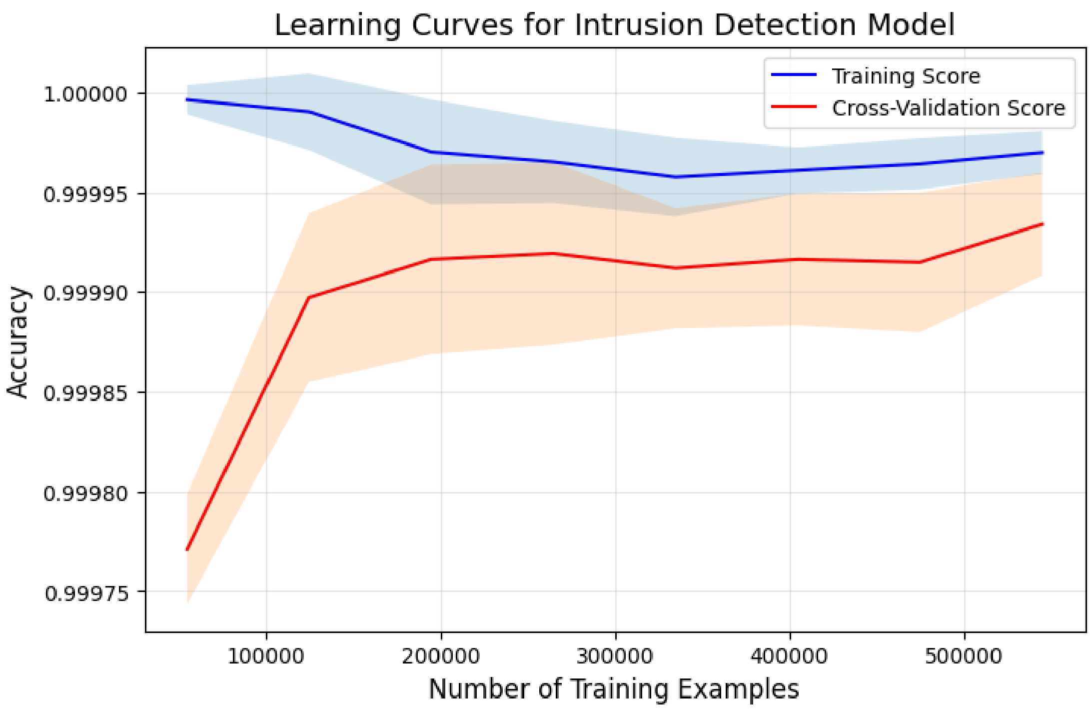
Top feature: sbytes (Impact: 33.56%)

Second feature: rate (Impact: 16.59%)

Third feature: pkts (Impact: 15.05%)





Comparing with individual models on test set...

	Accuracy	Balanced Acc	Precision	Recall	F1	MCC	Kappa \
xgb	1.0000	0.9924	1.0000	1.0000	1.0000	0.9914	0.9914
Ensemble	0.9999	0.9962	0.9999	0.9999	0.9999	0.9849	0.9849
lgb	0.9999	0.9962	0.9999	0.9999	0.9999	0.9679	0.9676

	Specificity	Sensitivity	AUC
xgb	0.9848	1.0000	0.9999
Ensemble	0.9924	1.0000	0.9981
lgb	0.9924	0.9999	0.9963

McNemar's test (Ensemble vs Logistic Regression): p-value = 0.000000

Result: Statistically significant difference ($p < 0.05$)

Model saved to fast_ids_model.pkl

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FINAL CLASSIFICATION REPORT (Test Set)

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	precision	recall	f1-score	support
Benign	0.98	0.99	0.98	527
Malicious	1.00	1.00	1.00	291345
accuracy			1.00	291872
macro avg	0.99	1.00	0.99	291872
weighted avg	1.00	1.00	1.00	291872

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INTRUSION DETECTION SYSTEM - COMPLETED SUCCESSFULLY

Results saved in: ids_results/

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Sample predictions on test instances:

Correct | True: Malicious | Predicted: Malicious

Correct | True: Malicious | Predicted: Malicious

Correct | True: Malicious | Predicted: Malicious

Correct | True: Malicious | Predicted: Malicious

Correct | True: Malicious | Predicted: Malicious

Total execution time: 9.65 minutes

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Top 10 features:

	Feature	Importance
0	delta_time	1236
1	port_src	585
2	port_dst	322
3	frame_len	304
5	ip_ttl	249
7	tos	200
6	icmp_type	100
4	udp_len	2
16	tcp_flags_ack	2
9	ip_flags_df	0

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SHAP analysis (sampling 100 test samples)...

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FEATURE IMPORTANCE BASED ON SHAP
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Feature                  Strength     Impact %
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frame_len                  3.2581      47.41
port_dst                   1.3421      19.53
tos                        0.9327      13.57
port_src                   0.7628      11.10
ip_ttl                     0.3880       5.65
icmp_type                  0.1389       2.02
delta_time                 0.0479       0.70
tcp_flags_ack              0.0019       0.03
udp_len                    0.0000       0.00
ip_flags_df                0.0000       0.00
ip_flags_mf                0.0000       0.00
tcp_flags_res              0.0000       0.00
ip_flags_rb                0.0000       0.00
tcp_flags_ns               0.0000       0.00
tcp_flags_cwr              0.0000       0.00
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Top feature: frame_len (Impact: 47.41%)
Second feature: port_dst (Impact: 19.53%)
Third feature: tos (Impact: 13.57%)
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